ROBOTICS / RESEARCH PROGRESS

Geometry-Constrained Bowl Manipulation

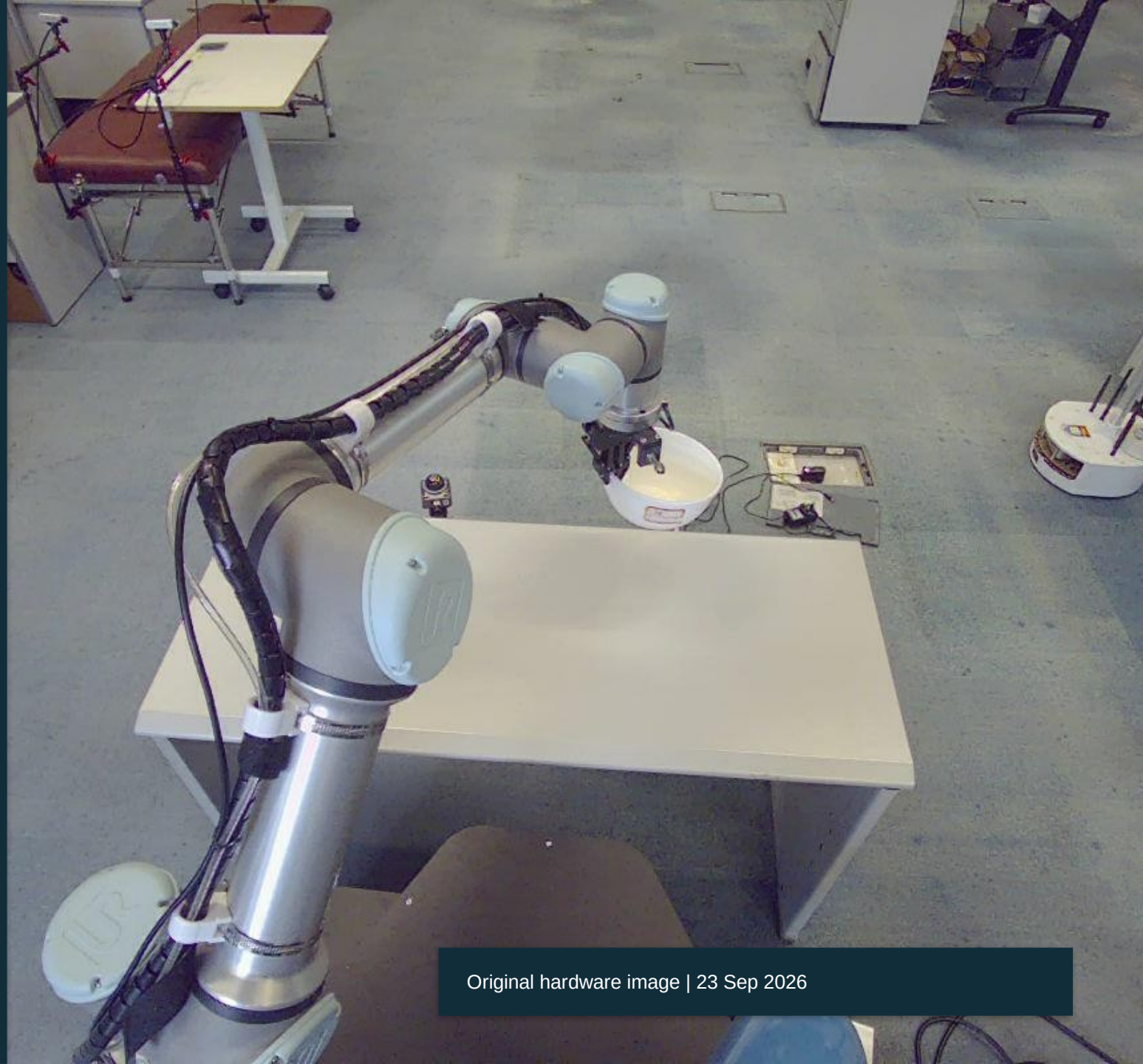
From visual rim grasping to repeatable placement

RGB-D perception

Radial contact geometry

Taught placement

02 OCT 2026



Original hardware image | 23 Sep 2026

A fixed tool pose does not fix the bowl center



AI-generated contact illustration | Same bowl type; different rim engagement

Contact geometry matters

Different contacts change the bowl-to-gripper offset.

Teaching controls the tool

A fixed release pose does not directly constrain the bowl center.

Design objective

Make the grasp relationship more consistent.

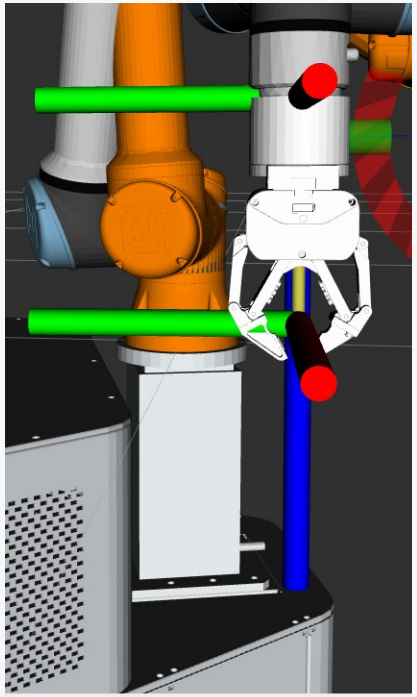
Principle illustration; no experimental measurements are represented.

Radial alignment with fixed finger assignment



Physical finger identification

Circled finger = left



RViz frame check

Green axis = tool +Y

Left outside; right inside

Preserve the identity of each physical finger.

Radial closing direction

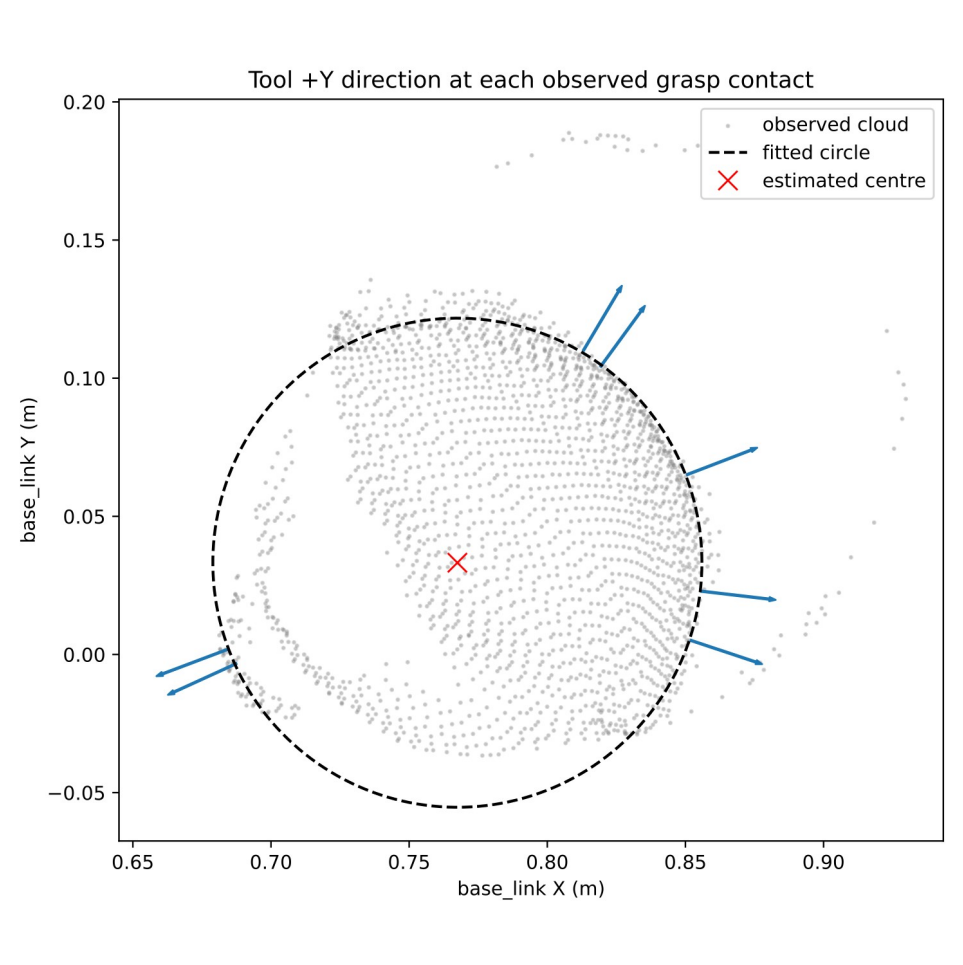
Tool +Y points outward; −Y points toward the bowl center.

Consistent engagement

Keep grasp height and insertion depth consistent.

Original user reference images; the physical left finger was identified as tool +Y.

Global shape guides, local geometry validates



OBSERVED GEOMETRY

Global shape

Fit the observed rim to estimate center and radial direction.

Local contact support

Check wall patches at multiple neighborhood scales.

Cross-check the grasp

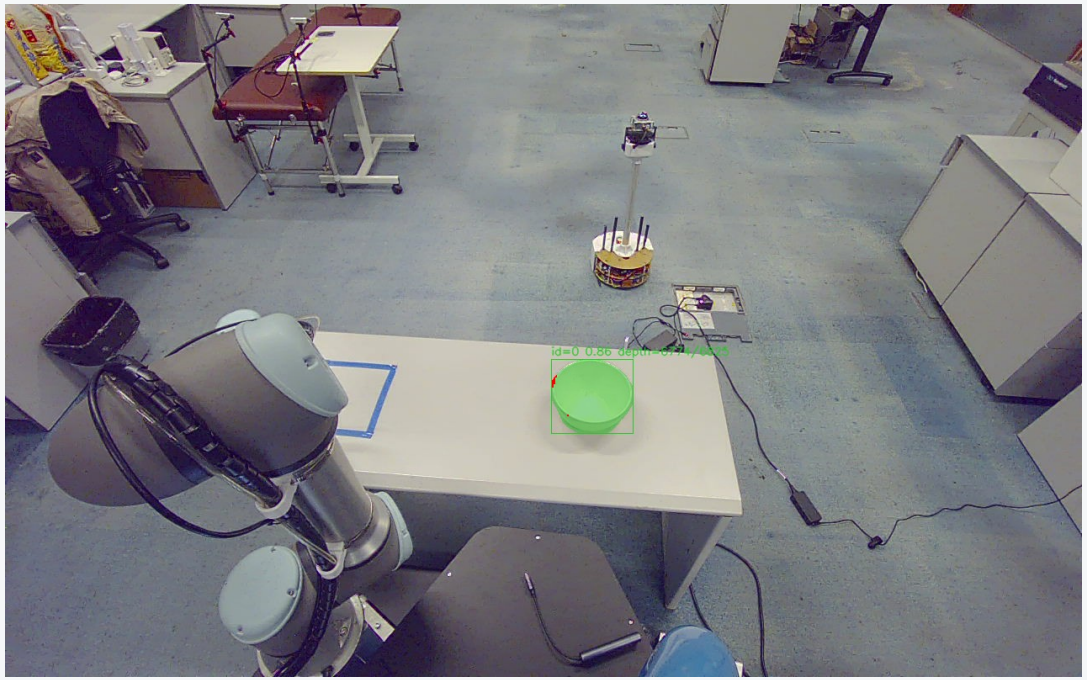
Compare the wall direction and observed contact with the circle.

Preserve measured positions

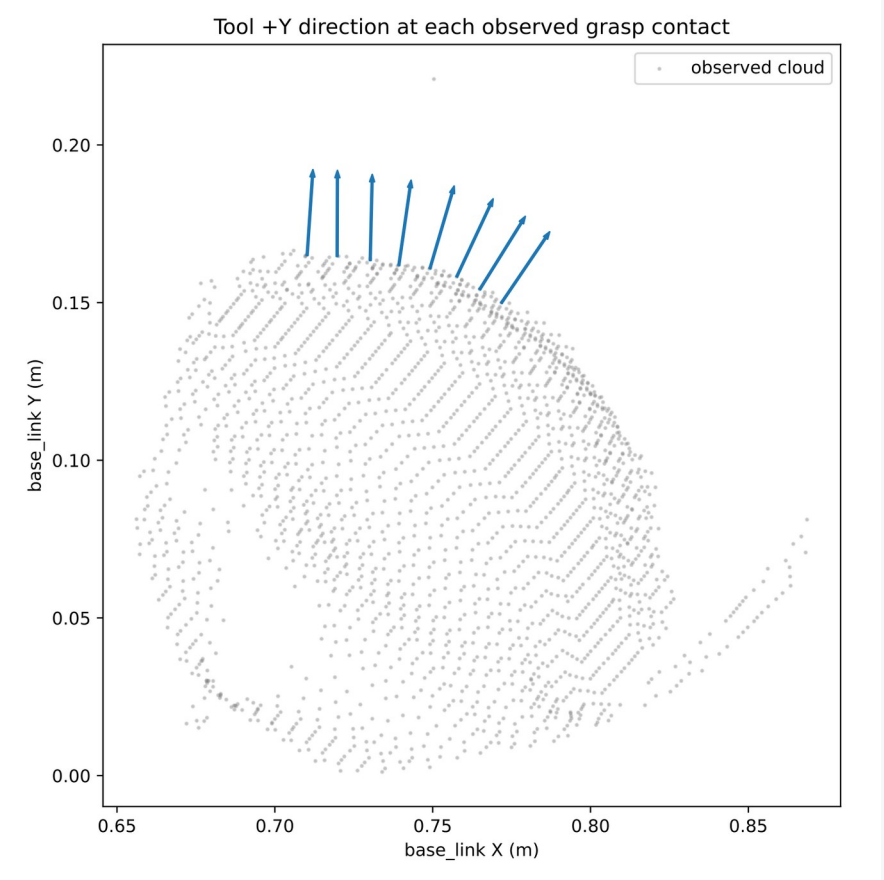
Do not project grasp contacts onto an idealized rim.

Original point-cloud replay visualization | circle_radial_replay_final_20260928

Use reliable local evidence when the rim is incomplete



LIVE PLANNING PREVIEW



Approximately 103° coverage; 8 grasp candidates.

Prefer validated radial geometry; use supported wall directions when arc coverage is insufficient.

Placement depends on the tool-to-center offset



Illustrative rendering; not a measured placement trial

BOWL CENTER AT RELEASE

$$c = p + R\,r$$

Bowl center = tool position + rotated object offset

Teaching fixes the release pose

Store full position and orientation in consistent frames.

Radial grasping constrains the offset

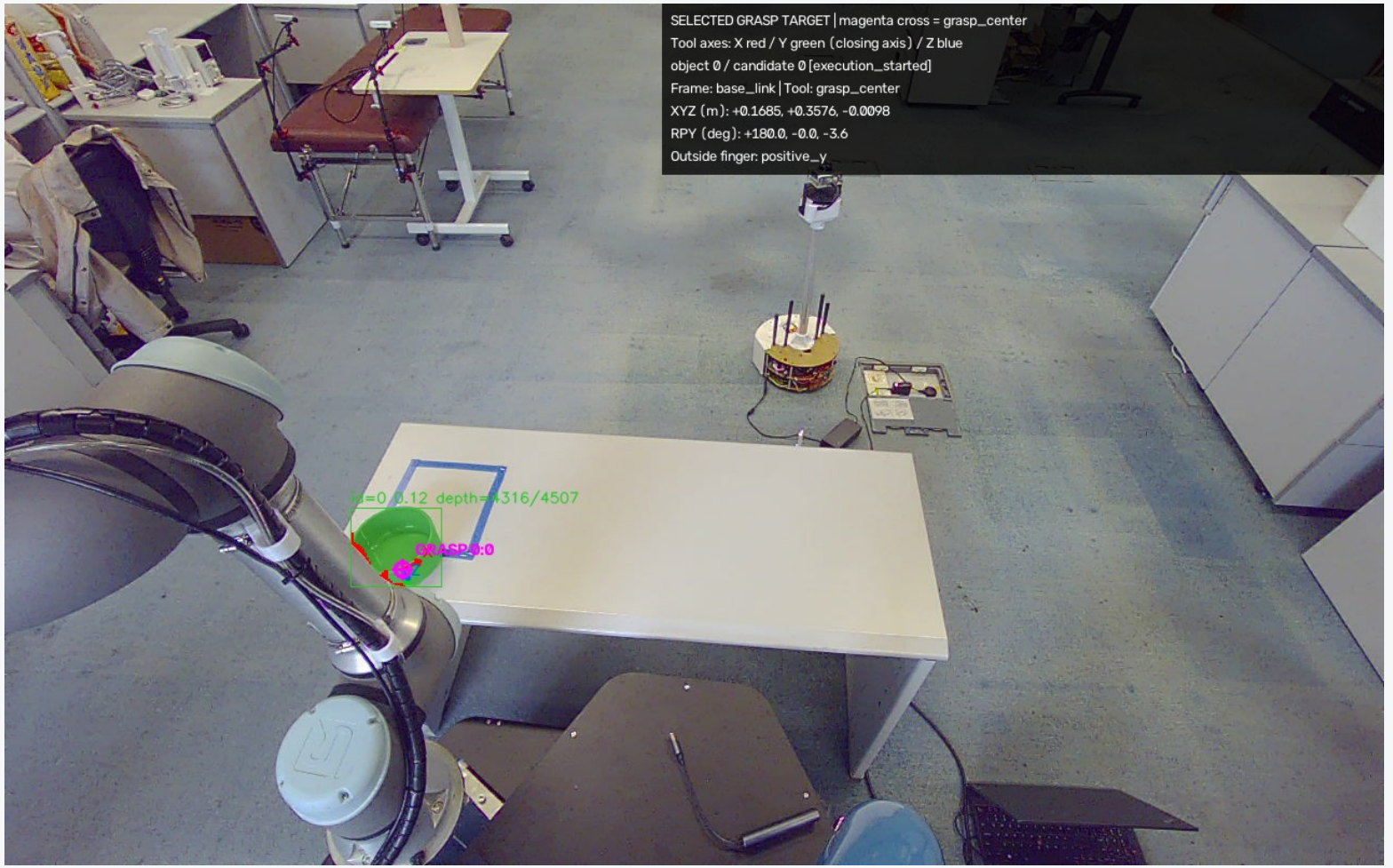
The aim is lower variation in the bowl-to-tool relationship.

Residual error remains

Calibration, contact depth and slip still affect placement.

Geometric explanation under fixed bowl geometry, consistent engagement, and no slip.

Select a grasp that supports the complete task



SEVEN MOTION SEGMENTS

01 Approach

02 Descend to grasp

03 Lift

04 Transfer to pre-release

05 Descend to release

06 Retreat

07 Return to saved home

Try another candidate before motion if complete planning fails.

Selected target overlay from a real task | 29 Sep 2026, 14:26:44 | The overlay is a target pose.

Advance execution using measured state



GRIPPER STATE CONFIRMATION

Confirm open before descent.

Confirm closed and settled before lift.

Confirm release before retreat.

Recorded feedback: Close 679 → 37 | Open 24 → 680

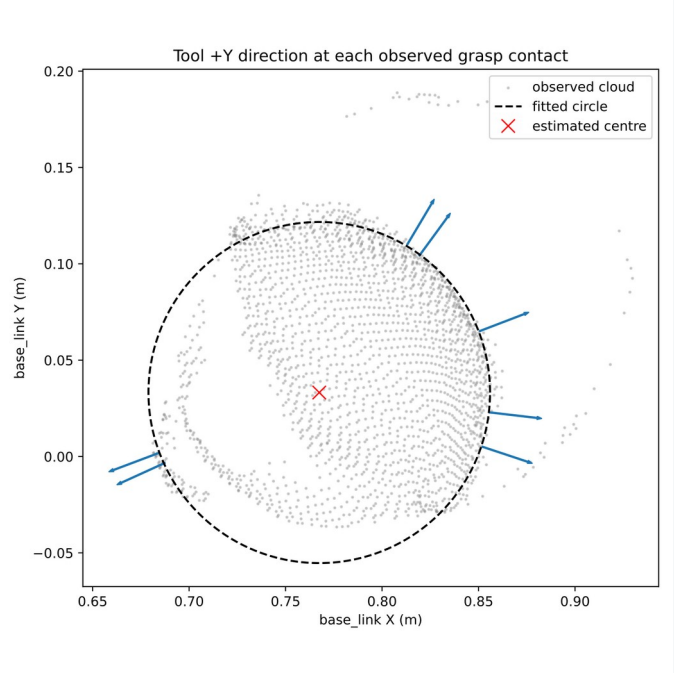
TEMPORARY TF LOSS

Use fresh joint measurements with validated forward kinematics.

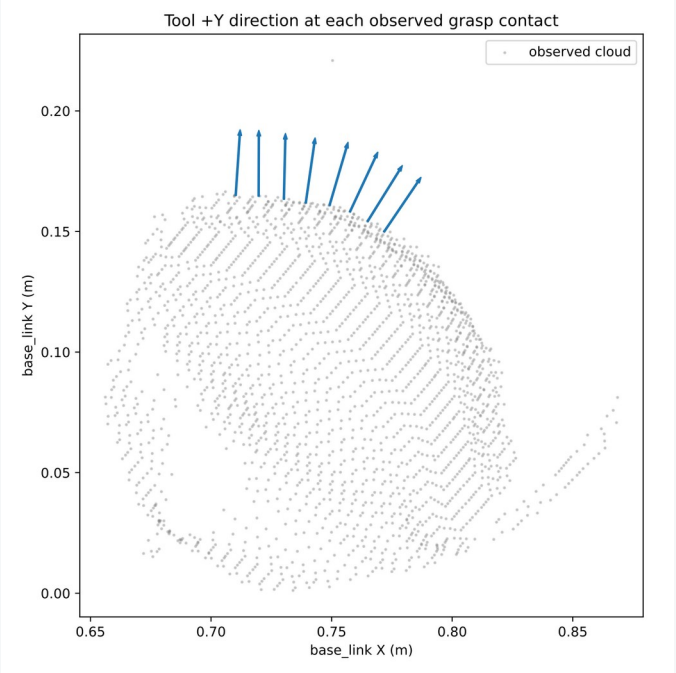
16 integration checks passed; 12 joint configurations cross-checked.

Gripper state confirmation does not verify that an object is held.

More observations produced valid grasp candidates



Observed-contact radial candidates



Supported local-wall fallback

FRAMES WITH VALID CANDIDATES

14 / 20 → 18 / 20

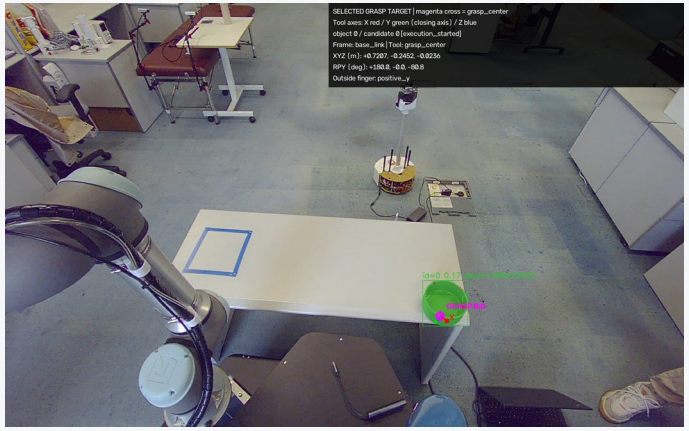
Partial-arc acceptance

23 / 27 → 27 / 27

Supported local-wall fallback

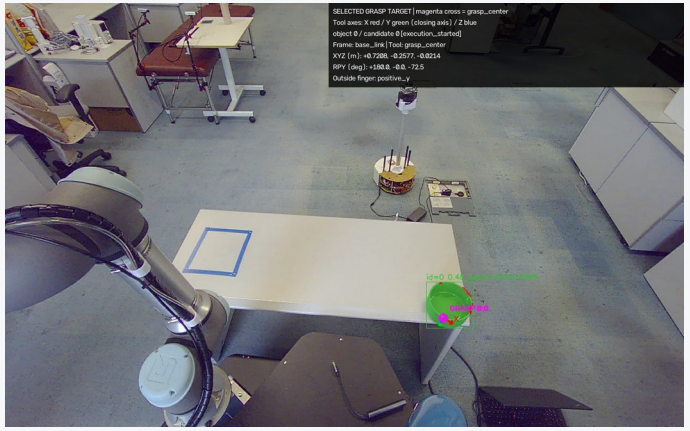
Two different replay sets. Candidate generation is not physical grasp success.

A complete single-bowl demonstration workflow



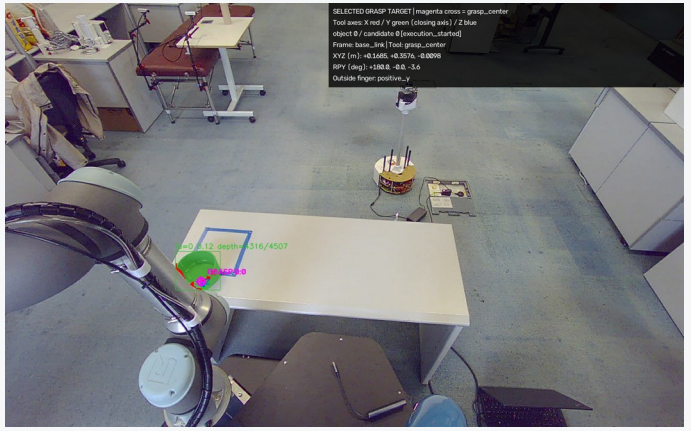
BLUE TARGET

14:20:46 | command sequence completed



UPPER LEFT

14:22:52 | command sequence completed



LOWER RIGHT

14:26:44 | command sequence completed

One command, one bowl

Fresh observation; release, retreat and fixed home return.

Six taught destinations

Five spatial zones plus blue; explicit XYZ input is also supported.

Traceable results

Saved masks, candidate poses, point clouds and configuration snapshots.

Images from separate trials, 29 Sep 2026 | Completed command sequences; physical outcomes not automatically verified.

Measure placement repeatability directly



Target quantity: bowl-center distance to the taught target

PLANNED COMPARISON

Wall-normal grasping vs. signed radial grasping

Vary the source position

Repeat across positions and grasp directions.

Measure physical outcomes

Grasp success, XY placement error and cycle time.

Separate the methods

Report radial and fallback trials independently.

Original scene image; no measured placement-error distribution is available in the current evidence set.

DESIGN CONTRIBUTIONS

Make the grasp relationship more consistent.

01 **Consistent geometry**

Radial alignment and fixed finger identity.

02 **Observed contact support**

Global shape with local wall validation.

03 **Complete-task feasibility**

Select candidates for the full motion cycle.

NEXT MILESTONE

Quantified physical success and placement repeatability



Original hardware photograph | 23 Sep 2026